

First-Order Linear Equations

MATH 146 – Calculus II

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A first order differential equation is said to be **linear** if it can be written in the form $y' + p(x)y = f(x)$.

Example: $x \frac{dy}{dx} = 4y + x^5 e^x$.

An equation of this form can be solved by using an *integrating factor*.

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Theorem

Given a first order differential equation written in the form $y' + p(x)y = f(x)$, the solution $y(x)$ to the equation is

$$y(x) = \frac{\int e^{\int p(x) dx} f(x) dx + C}{e^{\int p(x) dx}}$$

Example: $x \frac{dy}{dx} = 4y + x^5 e^x$.

Example: First Order Linear Equations

Example: $\frac{dy}{dx} + y = x, y(0) = 4.$

Example: First Order Linear Equations

Example: $y' = 2xy + 2$, $y(0) = 1$.

Example: First Order Linear Equations

Example: Find a general solution for the differential equation $xy' + 3y = 4x^2 - 3x$. Assume $x > 0$.